LiveTracker: Agentic AI for Behavioral Neuroscience Research

United Nations of UMB | Syed Taswar Mahbub, Ava Sajjadi, Igor Ten, Karla Schallies, Sadaf Hamidi| UMass Boston

Business Problem

- Behavioral researchers spend hours manually reviewing animal recordings
- Important behaviors, such as aggression, may last only seconds
- Manual scoring is time-consuming and prone to missed events
- Researchers need faster, more efficient ways to identify and investigate animal behavior

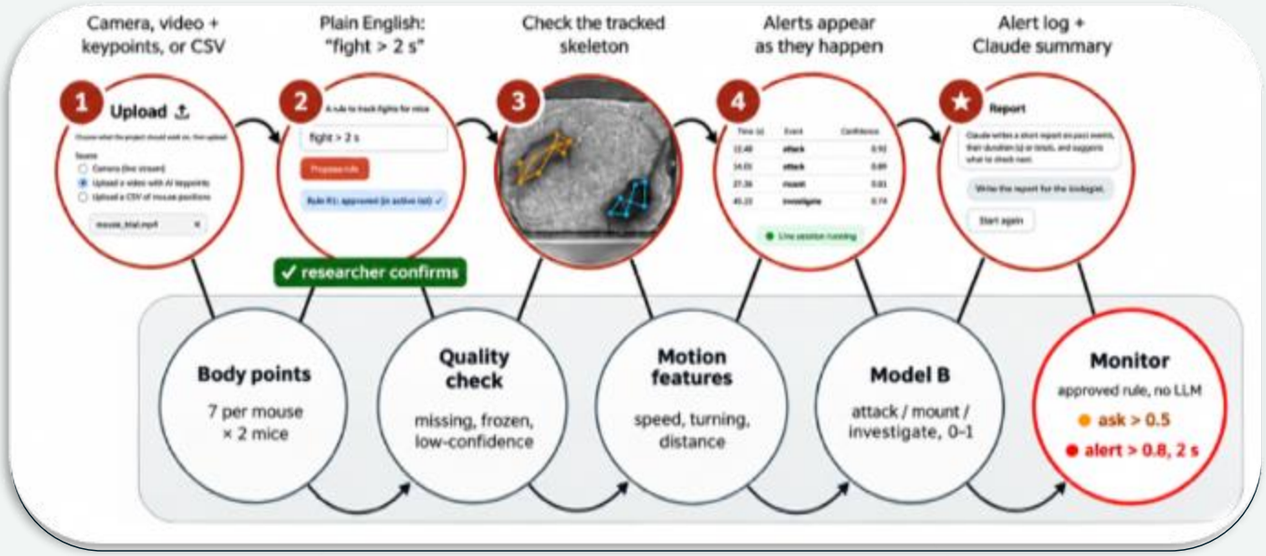
Target Users & Competitors

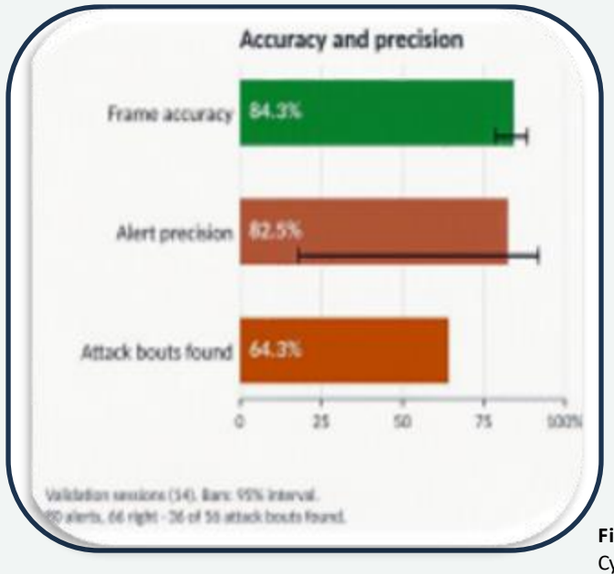
- Behavioral Neuroscience researchers
- Animal behavior laboratories
- Preclinical drug-testing teams

Competitors:

- Direct: MARS and SimBA, automated animal behavior analysis tools
- Indirect: Manual video scoring and traditional video-analysis methods

Application Workflow





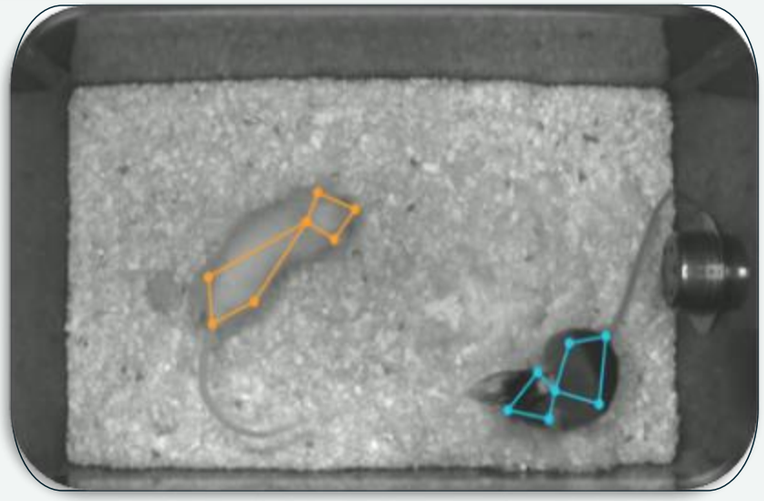


Figure 2: Live replay: mouse064 (CalMS21, frame 109)
Cyan = resident mouse, orange = intruder.
Solid dot = confident point · hollow = low confidence · cross = filled gap.

Key Features

- Detecting Behavior: predicts mounting, attack, and investigation
- Natural Language Rules: monitoring goals is described in plain English by researchers
- Smart Alerts: flags predicted events based on approved rules
- AI Investigation: Scientific tools are selected by an agent to examine behavioral patterns
- Evidence-Based Reports: Findings are linked to recorded measurements

Agent Evaluation & Responsible Use

Evaluations

- A total of 14 validation sessions containing 94,305 frames
- 83% attack-event precision
- 64% attack-event recall

Responsible AI

- Researcher approval required
- Findings supported by recorded evidence
- AI predictions ≠ confirmed behaviors

Sources & Acknowledgements

We thank Paul English Applied AI Institute (PEAAII) at UMass Boston for organizing our Applied AI Hackathon, our faculty mentors for their guidance, and to our judges for their time and feedback. We also thank the program's sponsors and funders for making this opportunity possible.

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